

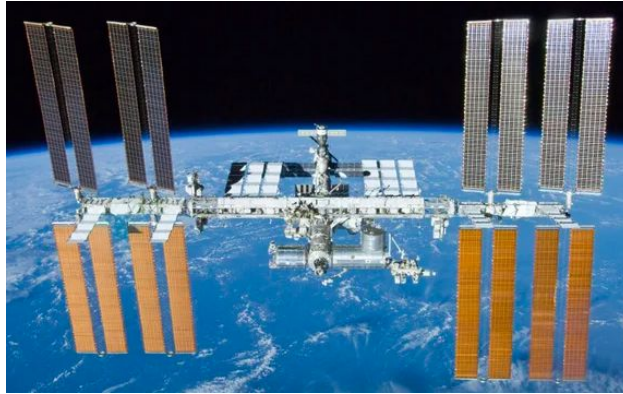
Let's learn about **satellites!**

Space mission and **satellite** design course

Satellite and space missions

What is satellite?

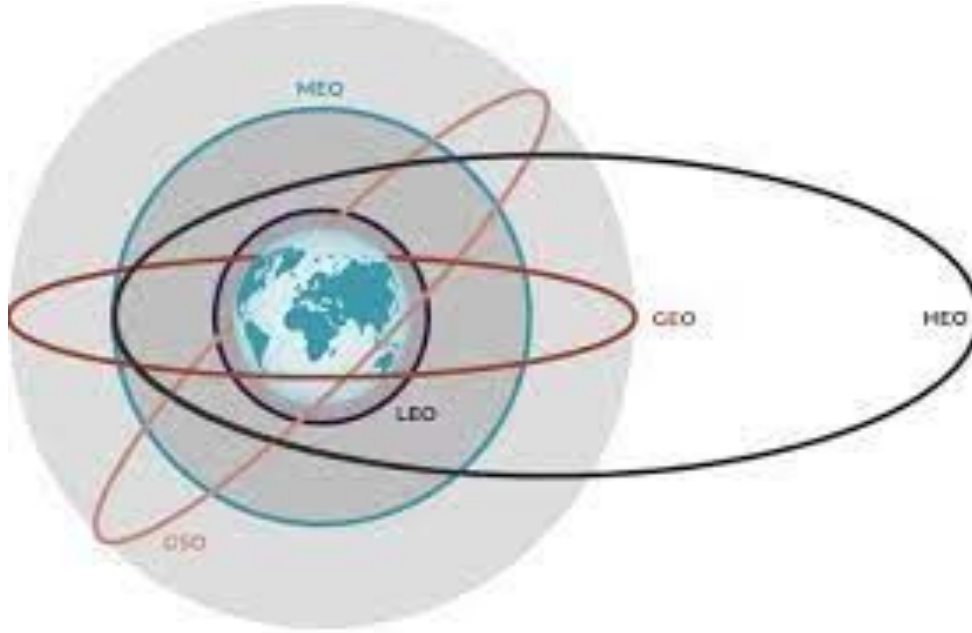
“A satellite or artificial satellite is an object intentionally placed into orbit around a celestial body”.



What is satellite?

- In the most common (and complete) way a satellite is a semi-autonomous system able to perform operations.
- This system (or object) is launched in space and placed into orbit around a celestial body, most often earth.
- An object in space can't fly, it is forced to move along an elliptical orbit in which the celestial body occupies one of the two foci.

What is satellite?



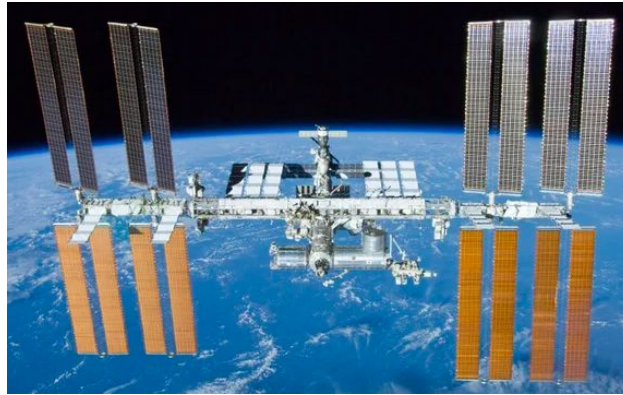
- Orbits are characterized by different parameters, determining geometry, position, distance from celestial object, etc..

What is satellite?

- Most of satellites are characterized by a relevant system called **PAYLOAD**
- The payload is in charge of performing the operation/scope/activity at the basis of the satellite construction and launch.
- The rest of the satellite at the service of the payload is commonly called **BUS**

Why do we launch satellites?

- A satellite is constructed and placed in orbit in order to perform several operation aimed to support **human life** on earth and beyond.



Question: what are these satellites doing for human life on earth?

Why do we launch satellites?

- The starting point is “the problem i am willing to solve”



Scope

The problem i need to solve (the data i need to collect, the service i need to provide)

Payload

The system/device that allows to perform the service, collect the data.

Satellite

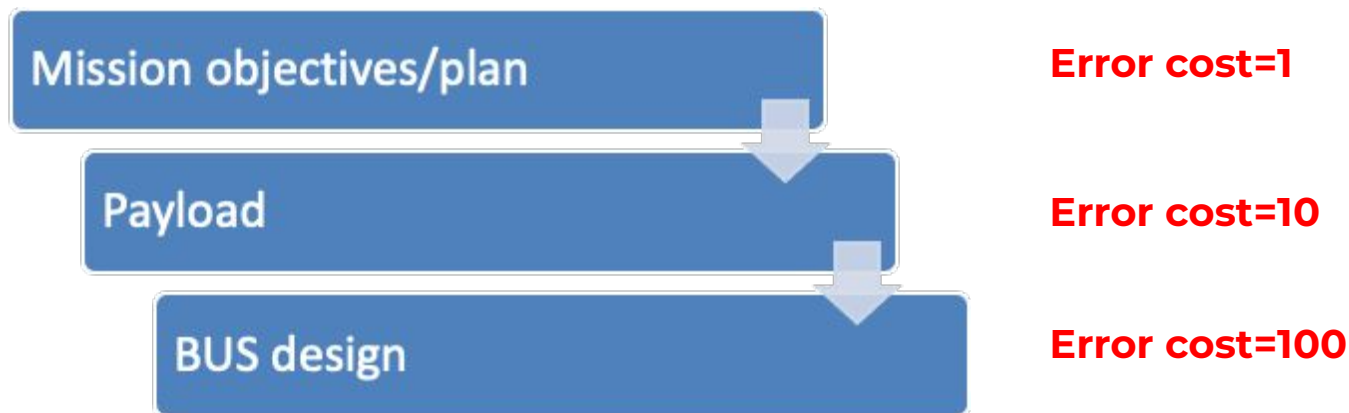
The vehicle, the system, that allows to bring the payload in space and operates

Satellite design

- The **design of the satellite** starts from definition of the **payload** requirements and constraints.
- The **payload design** starts from the definition of **mission** (scope) requirements and constraints
- The **mission** design starts from the understanding of the **problem** i need to solve and the way to pursue that (the data to be collected, the operation to be performed)

Satellite design

- An incorrect analysis upstream generates enormous complications downstream.
- The cost of misjudging payload requirements manifests itself in multiplied magnitude in subsequent stages!



Satellite classification

- Communications: make the connection through radio, television, and telephones.
- Weather: used for giving out continuous or up-dated pieces of information concerning the atmospheric conditions.
- Research: used for understanding the basic properties of materials and process in space, and to explore outer space, including features of celestial objects, magnetic fields, etc.
- Application: satellites used and designed to examine and test ways for improving the satellites technology of outer space.
- Navigational: satellites developed for navigation on earth
- Earth observation: satellite to perform remote sensing of earth
- Space observation: satellite to perform to perform space objet traffic management (space observation)

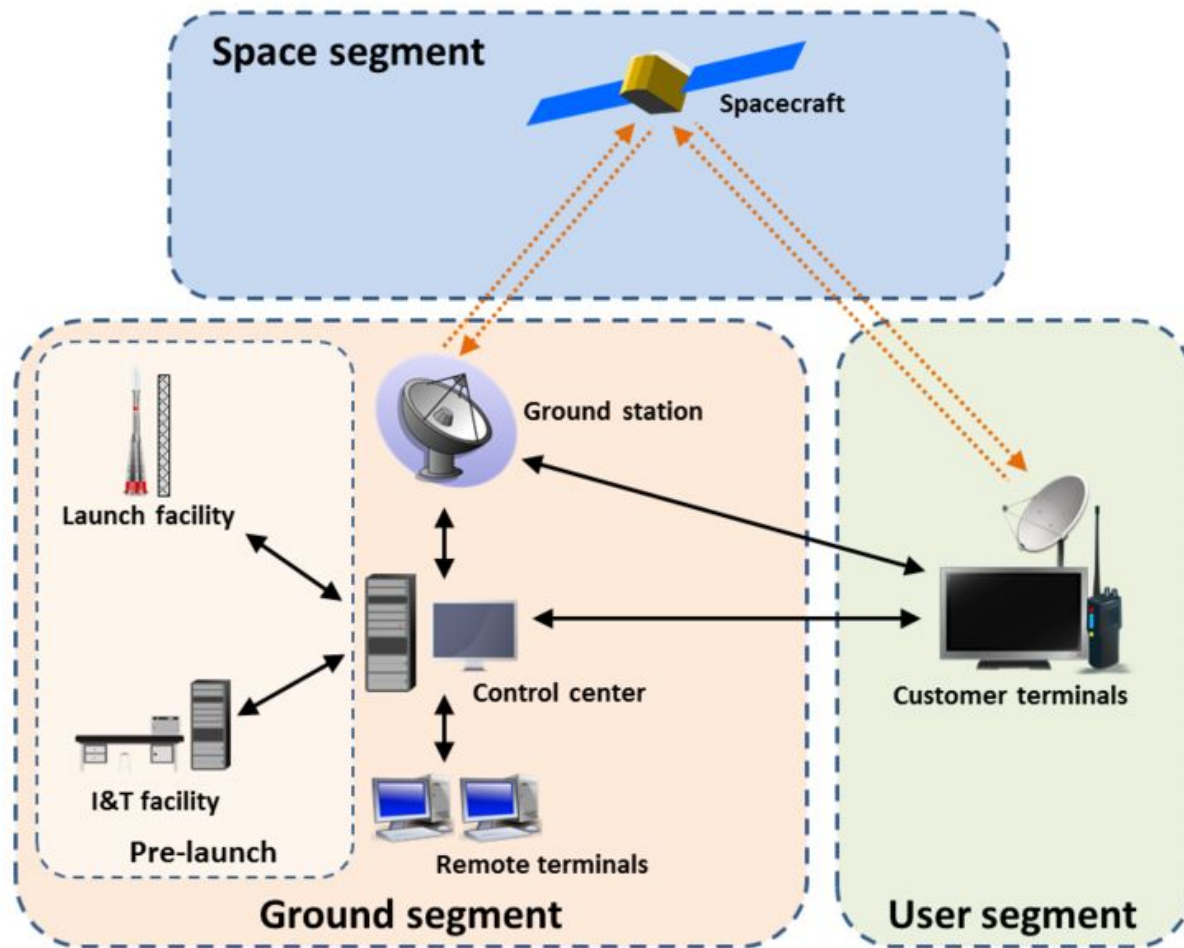
Satellite classification

Satellites are normally classified also in terms of mass:

- Large satellites: More than 1,000 kg.
- Medium-sized satellites: 500-1,000 kg.
- Minisatellite: 100-500 kg.
- **Microsatellite: 10-100 kg.**
- **Nanosatellite: 1-10 kg.**
- Picosatellite: Less than 1 kg.

What is a space mission

- By placing a satellite in orbit and performing operations we realize a **space mission**
- A space mission comprehend all the activities from the conception of the satellite up to the end of operation.
- Each space mission is characterized by 3 relevant segments



History of space

On Oct. 4, 1957, the Soviets launched the first artificial satellite, Sputnik 1, into space. Four years later on April 12, 1961, Russian Lt. Yuri Gagarin became the first human to orbit Earth in Vostok 1.



History of space



SPUTNIK-1 was a simple satellite (as per today state of the art) made by a sphere covered with radio transmitters.

SPUTNIK-1 transmitted for three consequent weeks until primary batteries run out.

History of space

- **Pioneer 10 and 11** 1972–2003
- **Venera probes** 1961–1984
- **Hayabusa** 2003–2010
- **Vostok 1** 1961
- **The Space Shuttle** 1981–2011
- **New Horizons** 2006–Present
- **Voyager 1 and 2** 1977–Present
- **Galileo** 1989–2003
- **Apollo 11** 1969
- **Hubble Space Telescope** 1990-present

Question: have you ever heard about these names?

Hayabusa-1



HAYABUSA the Asteroid Explorer

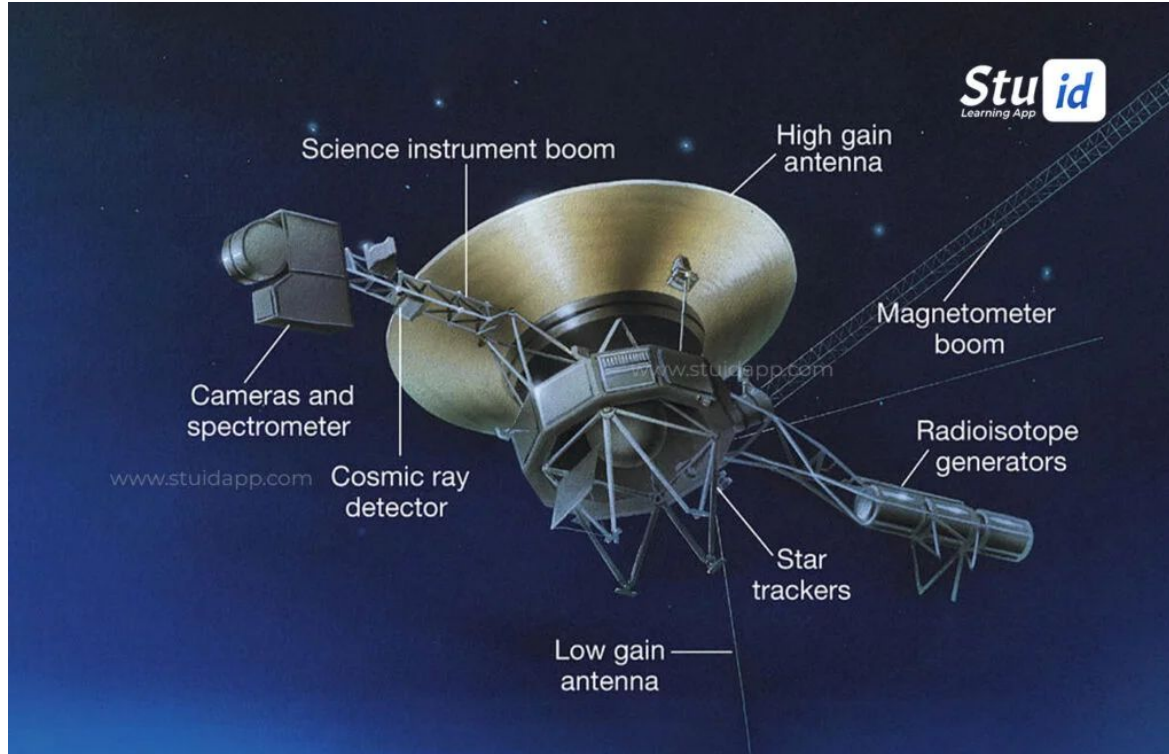
— 小惑星探査機「はやぶさ」とその成果 —



HAYABUSA

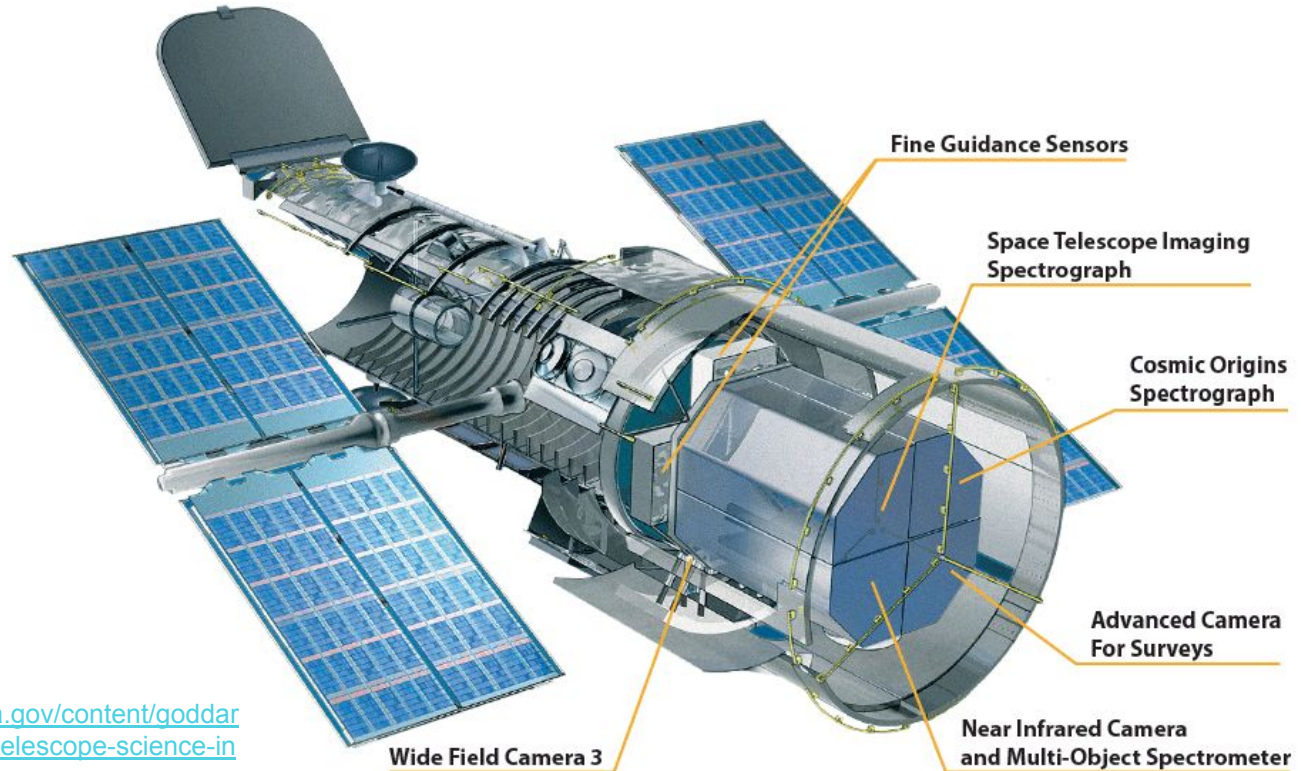


Voyager-1



<https://blog.stuidapp.com/the-mystery-of-voyager-1the-most-distant-human-made-object-in-space/>

Hubble space telescope



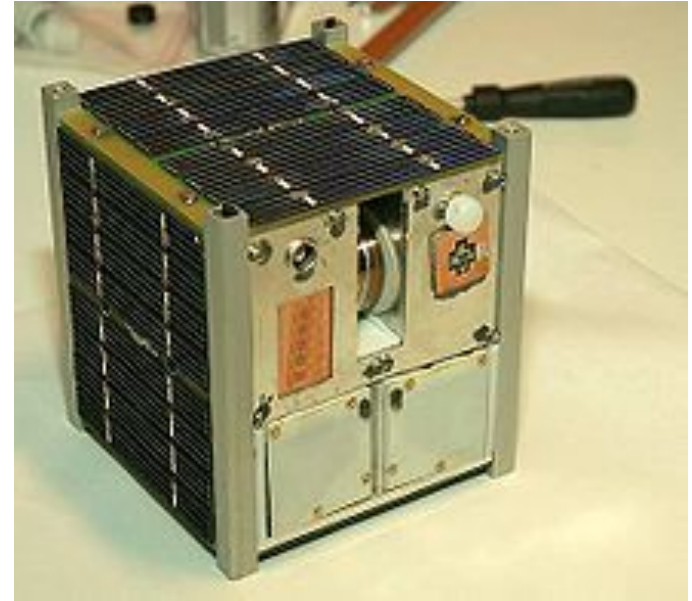
<https://www.nasa.gov/content/goddard/hubble-space-telescope-science-instruments>

Cubesat

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Cubesat

- The cubesat standard was introduced in 1999 by Prof. Jordi Puig Suari and Bob Twiggs.
- A Cubesat is a standardized nanosatellite class satellite characterized by a cubic shape of 10cm with a mass not higher than 2Kg. (1U)

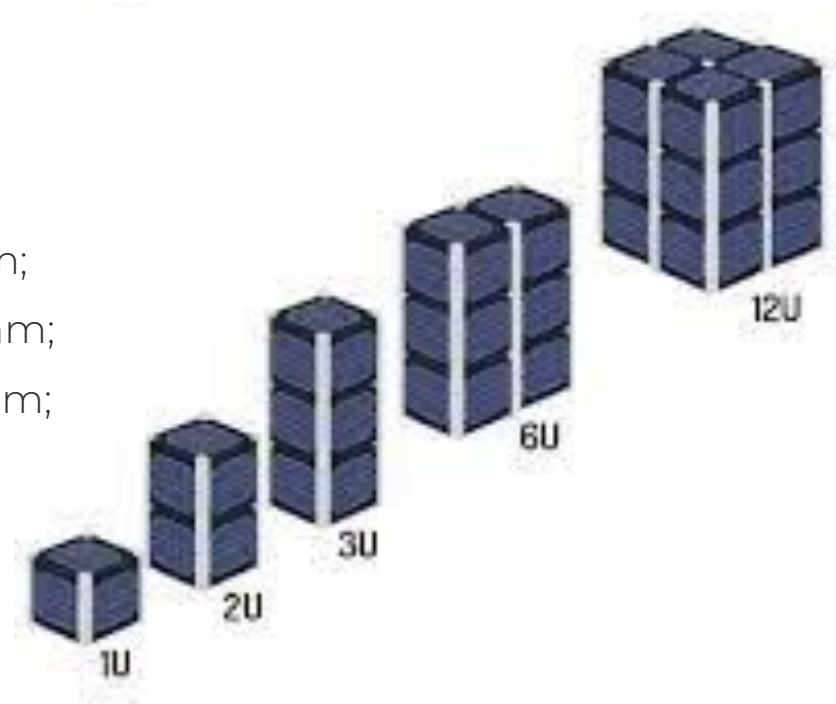


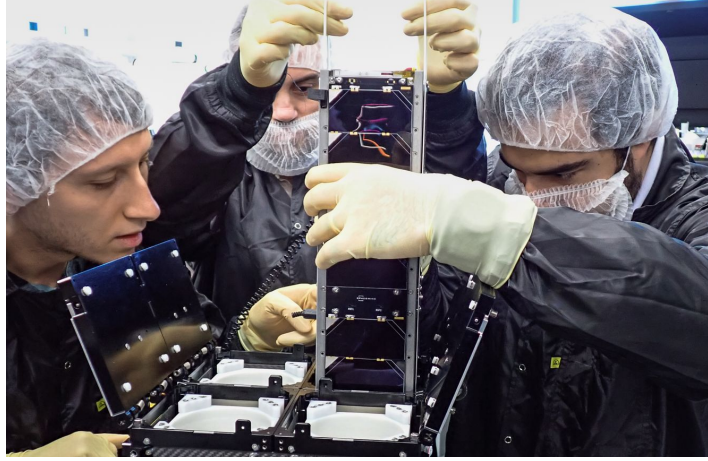
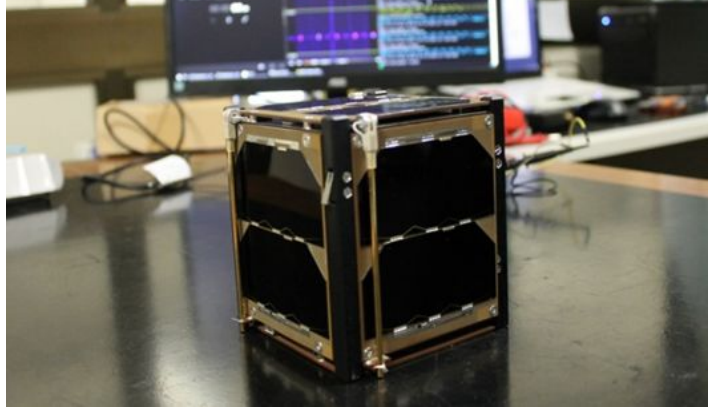
Cubesat

- Cubesats can be of different size always maintaining the concept of multiples of units (2U, 3U, 6U, 12U, 16U).
- Even if changing the size and weight, cubesats satellite are characterized by **standard interfaces and layouts**, outlined in the cubesat design specification from Calpoly.

Cubesat

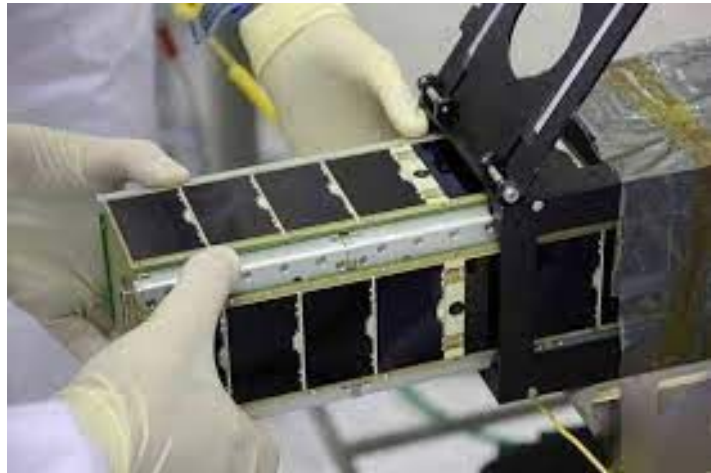
- 1U: 2 Kg - volume 100x100x100mm;
- 2U: 4 Kg - volume 100x100x227mm;
- 3U: 6 kg - volume 100x100x340mm;
- 6U: 12 Kg - volume 200x100x340mm;
- 12U: 24 kg - volume 200x200x340mm;
- 16U: 32 kg - volume 200x200x454mm;





Cubesat

- Cubesat standards introduced standardization in the deployment system (the device that allows detaching the satellite from the rocket in space)
- This system is the deployer: a jack in the box dispenser that ejects the satellite.



Cubesat



Like Cubesats, deployers are characterized by different sizes

Images: ISIS Innovative Solutions in Space cubesat deployers

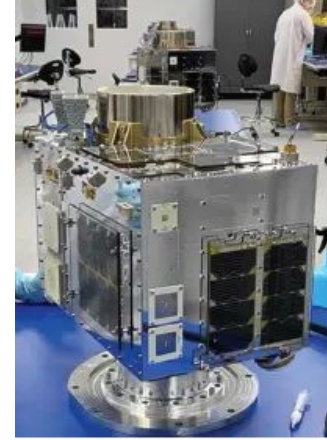
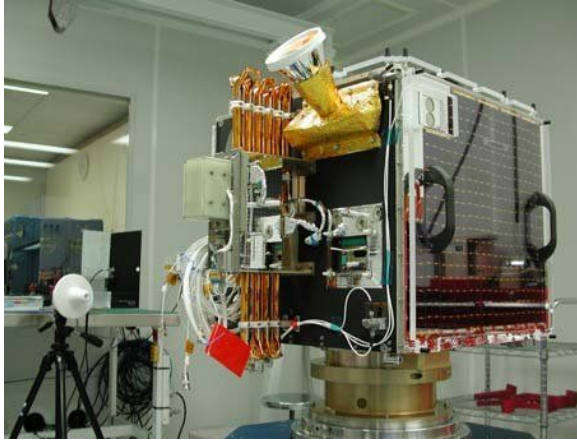
Cubesat

Question:

Why do we study Cubesat?

Why Cubesat are important?

Spot the difference - 1

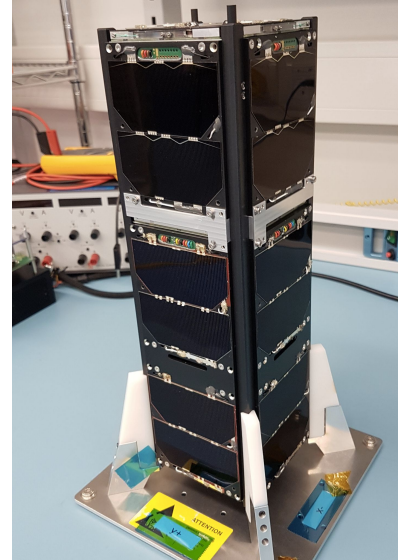
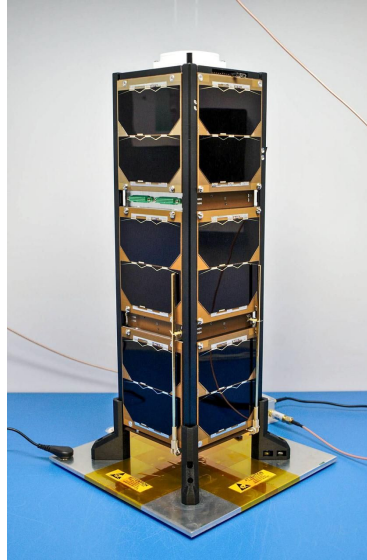


https://www.researchgate.net/figure/The-FedSat-satellite-during-integration_fig3_237643943

<https://www.utias-sfl.net/what-we-do/satellite-development-overview/>

<https://spacenews.com/atk-ships-ors-1-bus-prime-contractor/>

Spot the difference - 2



<https://www.nasa.gov/feature/elana-31-mission-cubesats-deployed>

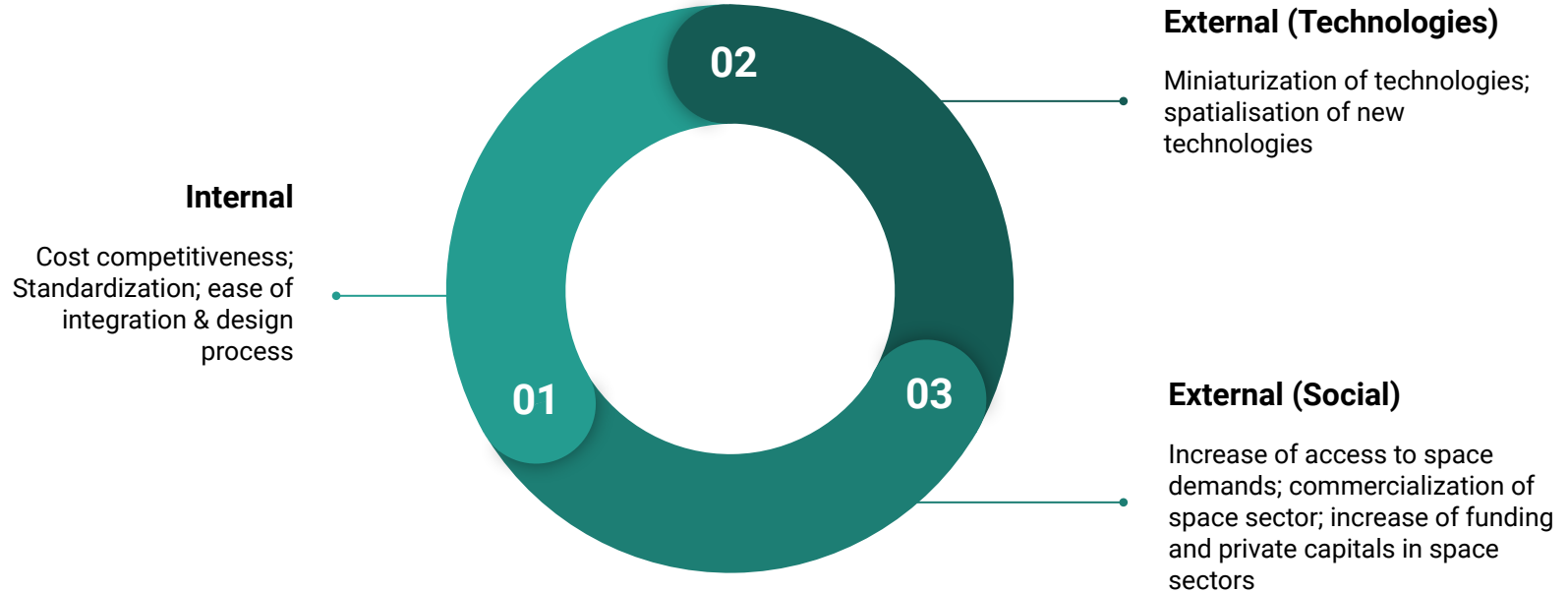
https://commons.wikimedia.org/wiki/File:The_RAX-2_Spacecraft.jpg

<https://spaceflight101.com/pslv-c37/pslv-c37-cubesats/>

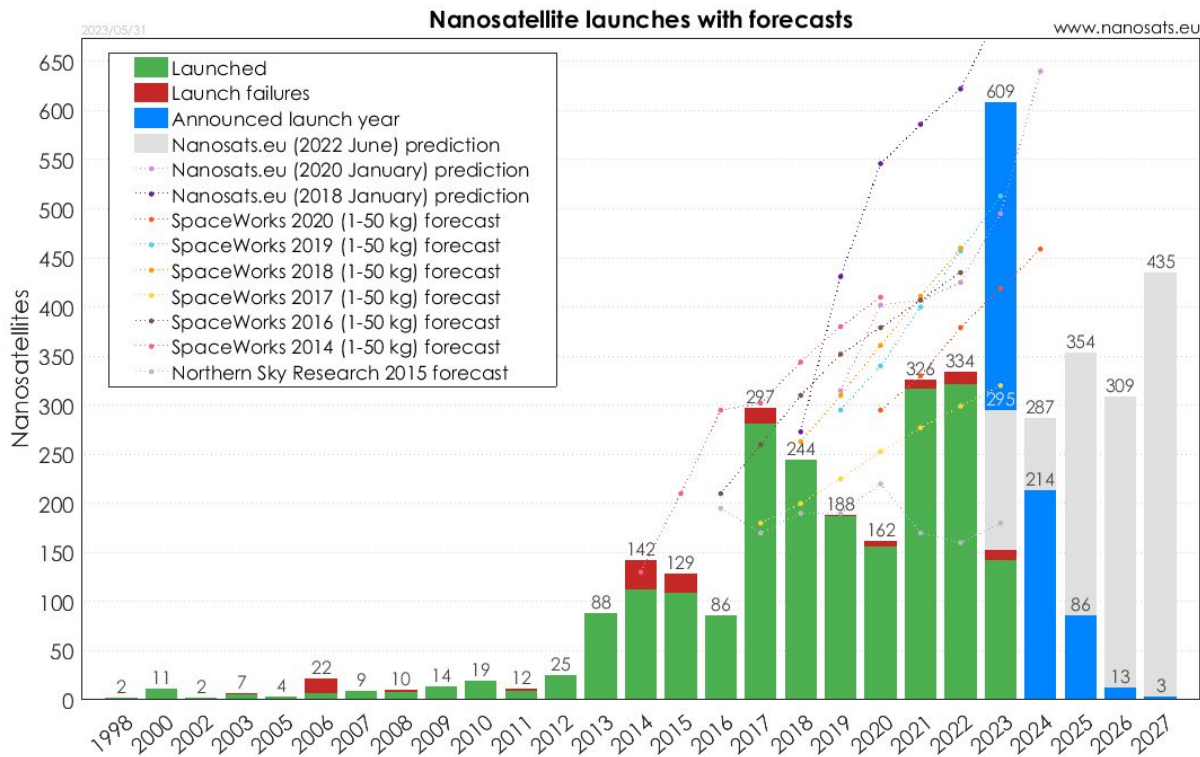
Cubesat

- By introducing **standardization** of geometry and interfaces, it has been given the possibility to companies to produce subsystems on the basis of shared requirements.
- This allowed the creation of **MARKET** with the birth of specific subsystem providers, platform providers, service providers.

Cubesat



Cubesat



<https://www.nanosats.eu/>